

FINAL SEMINAR · OCTOBER 2026

# Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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# Roadmap

## 01 Background

Multi-page, visually rich document understanding: what the documents are, what the task is, and how the field got here.

## 02 Survey

*Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding*

EMNLP 2026, main conference

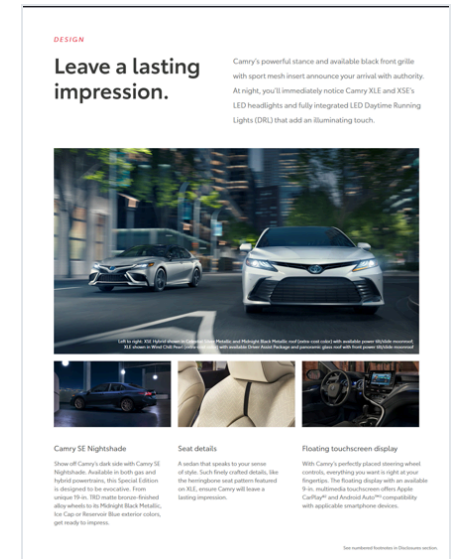
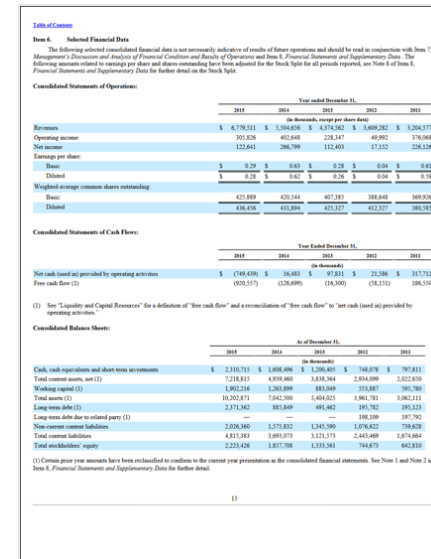
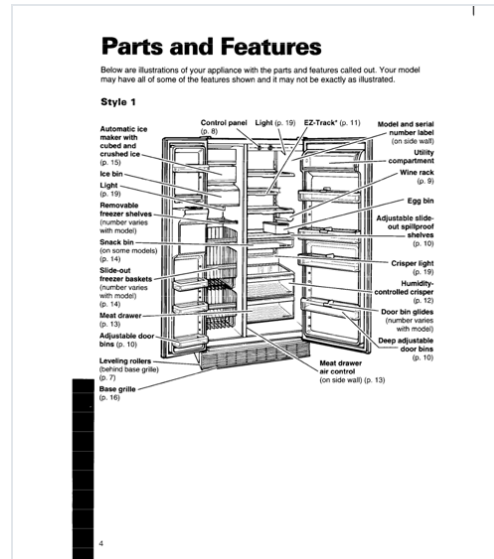
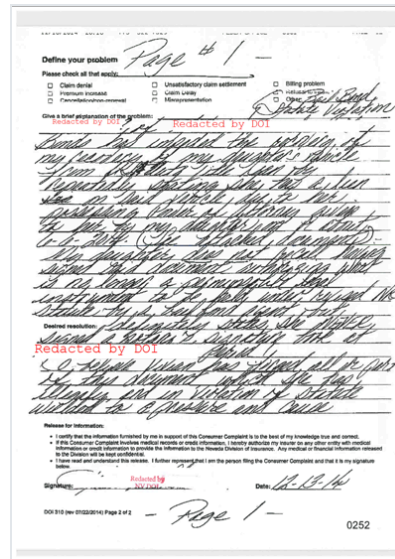
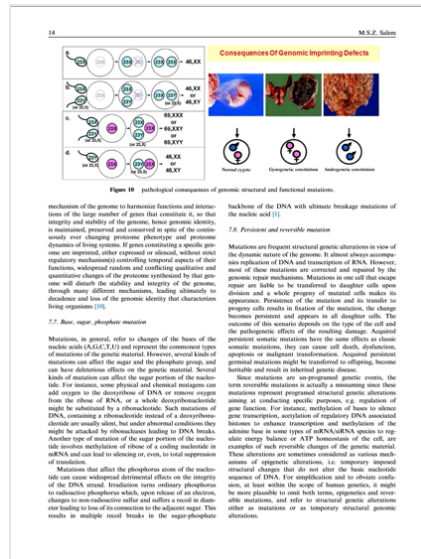
## 03 Empirical attribution study

*Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution*

Submitted to EACL, under review

# Multi-Page Visually Rich Documents

Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.



Research paper  
born-digital

Signed form  
scanned, handwritten

Technical manual  
layout-dependent

Financial report  
dense tables

Product brochure  
photo-led

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

# What is Document Understanding?

One broad way to define it: get a system to **answer a natural-language question over a document**, accurately. The systems range from rule-based lookup, through trained vision-language models, to pipelines built around pretrained general-domain ones.

- 1

"What year was this report published?"

FACT LOOKUP
- 2

"According to the table, which region had the highest revenue in 2023?"

READ A TABLE
- 3

"How much did the leading region's revenue grow between the 2022 and 2023 results tables?"

TWO TABLES, THEN ARITHMETIC
- 4

"Does the growth described in the executive summary agree with the figures in the results table?"

PROSE AGAINST NUMBERS
- 5

"Across the report and its appendix, what evidence supports the conclusion that the new method improved performance, and what limitation qualifies it?"

GATHER, THEN QUALIFY

Illustrative questions, not quoted benchmark items

# A Brief History

A useful starting point is **MP-DocVQA** in 2023: a benchmark of documents up to twenty pages, with questions answerable from a single page, plus the Hi-VT5 baseline. It is the pivotal moment at which long-document understanding became its own problem.

## BENCHMARKS

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### MP-DocVQA, 2023

Up to twenty pages. One page carries the answer, so the task is finding it.

### MMLongBench-Doc, LongDocURL

- Documents into the hundreds of pages, with questions that combine evidence from several places at once.

## MODELS

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### Purpose-built, trained end to end

Hi-VT5 and its successors: a document encoder trained for this setting specifically.

### General-domain VLMs as the backbone

- Increasingly sophisticated training and training-free machinery built around a pretrained model that already reads text and page images well.

A lot happened in between, and quickly. Organising that is what the survey does.

# Document Questions in Consumer Assistants

Everyone here has used these. What changed is not the question being asked, but **how much of the document the model has to hold at once**.

A YEAR OR TWO AGO

## The whole document, in context

Parsed text plus an image of every page, all placed in the model's context window. Fine for short documents, and it hits the ceiling fast.

SCREENSHOT  
"cannot upload document,  
out of context" error

MORE RECENTLY

## On disk, read by tool calls

The upload lands in the model's working environment. It reads what it needs, when it needs it, rather than ingesting everything up front.

SCREENSHOT  
assistant reading an uploaded  
file through a tool call

AND NOW

## Projects, searched by retrieval

Upload many documents at once. Beyond what fits in context, the content is indexed and retrieved against the question instead.

SCREENSHOT  
a project with several  
uploaded documents

# Real-World Constraints

A hospital, a mining company, or a law firm wants the same thing: **classification, summarisation, or structured extraction over tens of thousands of documents**. The consumer path is often not available to them.

## WHY NOT JUST CALL AN API

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- **The documents cannot leave.** Medical records, legal and financial filings, and resources data carry privacy and data-residency requirements that rule out a third party.
- **The volume makes it uneconomic.** At tens of thousands of documents, per-call pricing stops being worth the benefit even where sending the data is permitted.
- **The document type is unusual.** A niche form or report may need its own training or fine-tuning rather than whatever a general model already knows.

## WHAT THAT FORCES

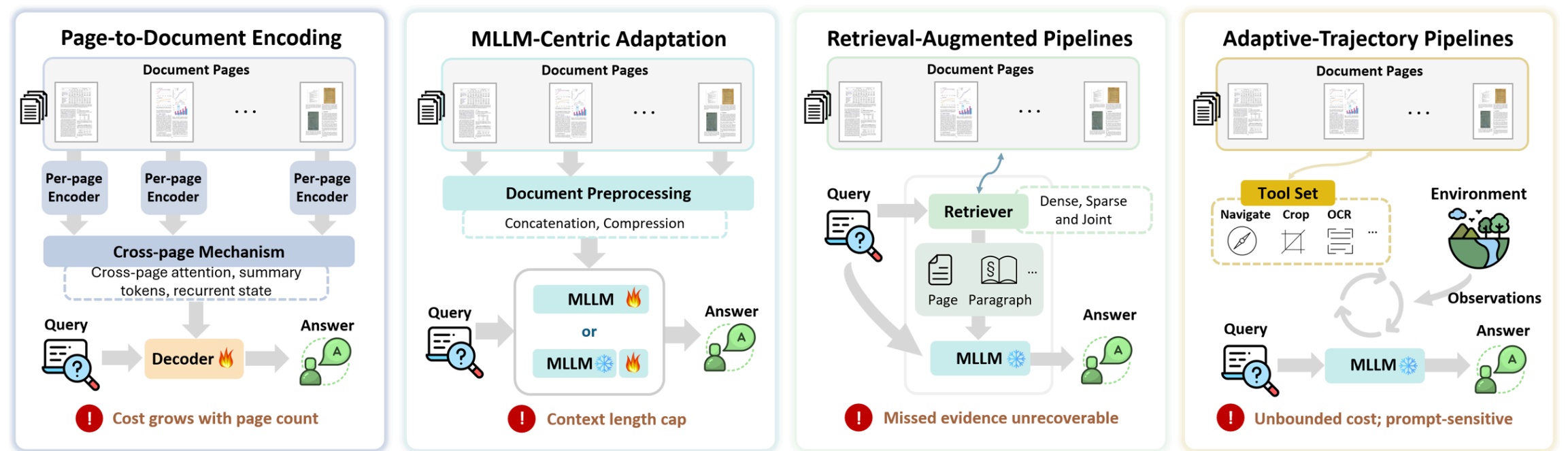
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- **Open-weight models, run locally.** A pretrained vision-language model on consumer hardware, not a datacentre.
- **A fixed compute budget.** Which pages are read, at what resolution, and how many times, all have to fit inside it.
- **Efficiency as a research problem.** Cost and latency are open questions in their own right, not implementation details.

This is where the research sits: **techniques and architectures for document understanding under a fixed compute budget**. The same techniques then flow back into the consumer products, which is why those have moved so quickly.

# Multi-Page Architectural Overview

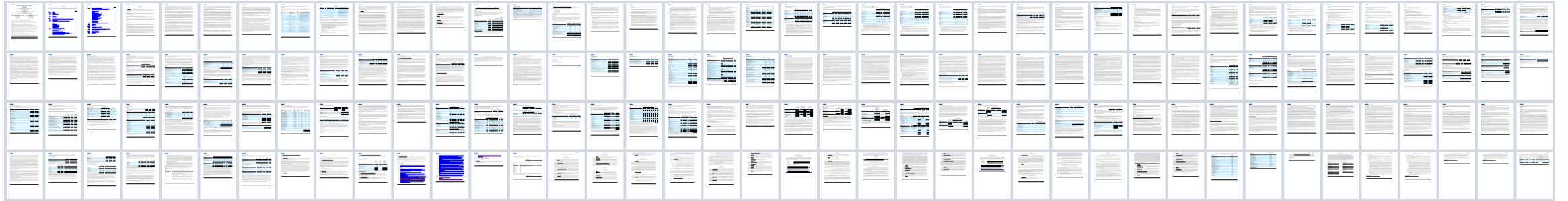
The survey groups systems by **how they manage evidence across pages**. One family encodes the pages together; one adapts a multimodal model to take more pages directly; one retrieves relevant pages before generating; one inspects pages step by step, deciding what to look at next.



Snowflake indicates typically frozen, fire indicates typically trained

# Text Modality

**VOLUME** one annual report: 160 pages, about 89,000 words



**CONTINUITY** a sentence cut by the page

Despite the limitations of our work, we find our results to be highly encouraging. We show that substantial weak-to-strong generalization is not only possible, but actually a widespread phenomenon. We also show that with very simple methods, we can drastically improve the ability of weak supervisors to elicit knowledge from strong models. With much more progress in this direction, we could get to the point where we can use weak supervisors to reliably elicit knowledge from much stronger

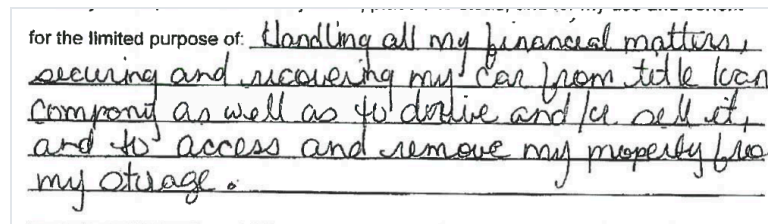
3

PAGE 3 ENDS, PAGE 4 STARTS

models, at least for some key tasks that we care about. This may allow us to develop superhuman reward models or safety classifiers, which we could in turn use to align superhuman models.

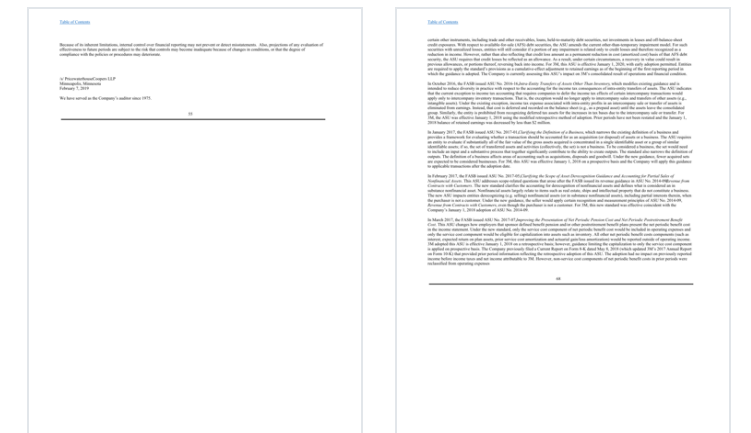
Aligning superhuman models is essential for making them safe; there is increasing recognition that failing to align such powerful models has the potential to be catastrophic, making this one of the most important unsolved technical problems in the world (CAIS). We think it is now more tractable than ever to make rapid iterative empirical progress toward solving this problem.

**NOISE** what OCR reads back



for the limited purpose of: \_ Liane in OA Wy J A anual mallins  
 QLEUIINA OO DALCOLON IDO Wi 'On LfLiorv AS iC  
 mond OQ oN Oo Yo! OWE end |e ud M\_ Lk  
 AY RACDAS GIA AMM Ore noah were rly f-As0-,  
 ea} OL O60

HETEROGENEITY same report, two pages



p.55 · 75 words

p.68 • 946 words

# Visual Modality

SIGNAL COMPETITION one page, one budget

3  
PEW RESEARCH CENTER

### Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

*Yet many economic indicators show few gains for the community since the Great Recession*

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the recession.

**PROSE**

Four-in-ten Latinos say their personal finances are in "excellent" or "good" shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007).<sup>1</sup> By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (41% in 2008 vs. 43% in 2015).

The survey also shows that Hispanics are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (81%) say they expect their family's financial situation to improve in the next year, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat –

**CHART**

**Hispanics more confident in personal finances than in 2008, see better economic times ahead**

% who say their personal financial situation is in excellent/good shape

% who expect their family's financial situation to improve "a lot" or "some" in the next year

| Year | Latino | General public |
|------|--------|----------------|
| 2004 | 31     | 51             |
| 2008 | 23     | 41             |
| 2011 | 24     | 38             |
| 2015 | 33     | 43             |

| Year | Latino | General public |
|------|--------|----------------|
| 2004 | 70     | 76             |
| 2008 | 67     | 56             |
| 2011 | 67     | 58             |
| 2015 | 81     | 61             |

Source: National Survey of Latinos conducted Oct. 21-Nov. 30, 2015; surveys conducted Dec. 8-13, 2015; Oct. 24-28, 2012; Sept. 12-16, 2012; Dec. 7-11, 2011; Dec. 3-8, 2008; Oct. 9-12, 2008; Aug. 5-10, 2004.  
1. Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

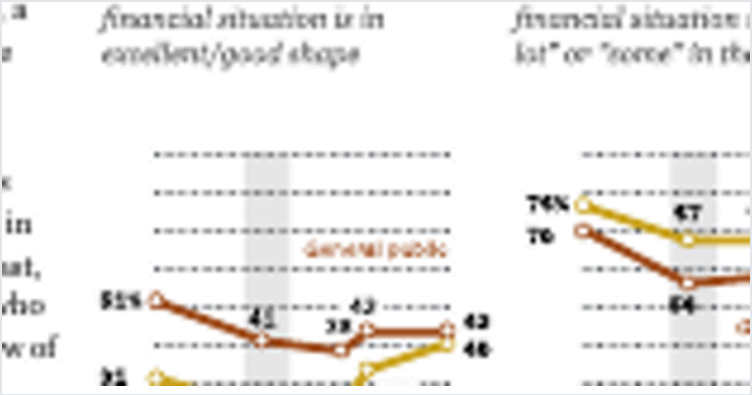
PEW RESEARCH CENTER

**FINE PRINT**

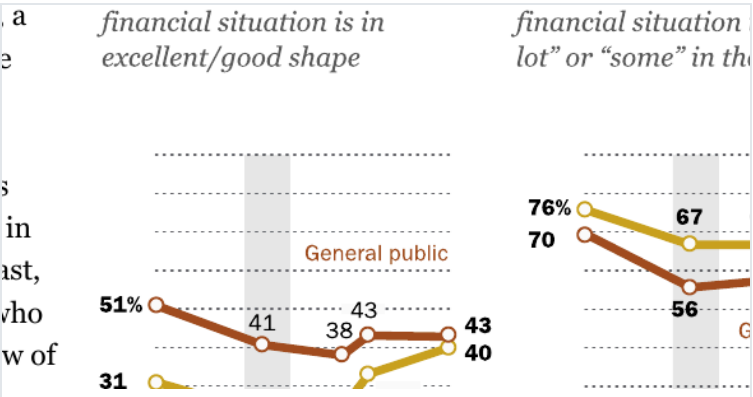
<sup>1</sup> The Great Recession lasted from December 2007 to June 2009, according to economists at the National Bureau of Economic Research.

www.pewresearch.org

RESOLUTION the same chart, twice

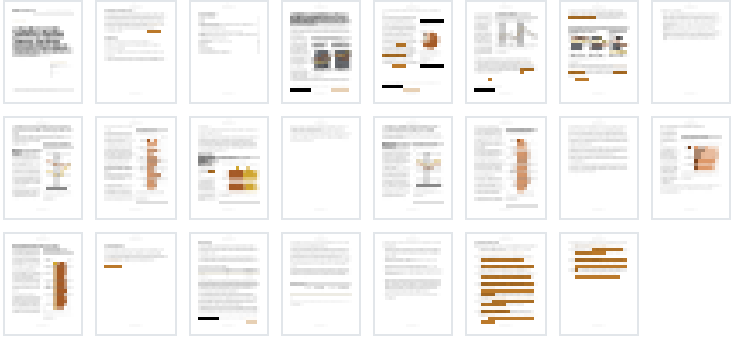


Page at 340 × 440 px · labels gone

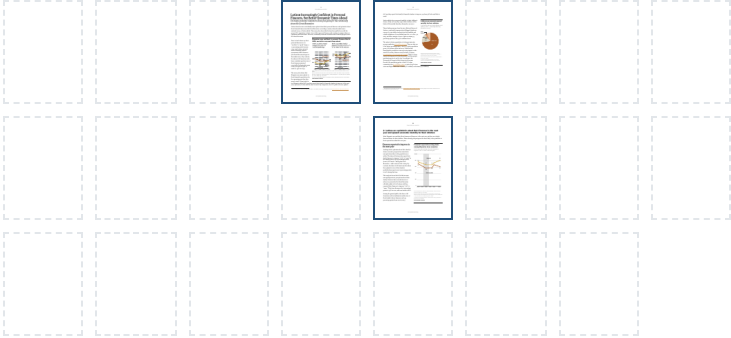


Page at 1275 × 1650 px · 14 times the pixels

COMPRESS OR SELECT 23 pages, same budget



Compress · every page, none of them readable



Select · three readable pages, twenty never seen

**HIERARCHY** which section a page sits under

**Survey report** • two chapters, each with its own subsections

prehension by both the models and humans, since WikiData utilizes QIDs (e.g., Q220) to represent unique entities. We construct a prompt (Table 13 in appendix D) which includes 1) instruction to the models to generate attributed answers. 2) retrieved knowledge graph, and 3) the question. We

The sentence is on page 4. Table 13 is fifteen pages later, in an appendix.

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| <p>topics, trends, and issues, building on others' ideas and expressing their own clearly and persuasively.</p> <p><b>Speaking and Listening 12.1-2.2</b> Integrate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, and verbally) in order to make a case on a topic or issue, evaluating the credibility and accuracy of each source and noting any discrepancies among the data.</p> <p><b>Reading for Informational Text 11.7-2.4</b> Analyze a complex text of several sequences of expository and/or argumentative texts in order to understand how the different parts of the text contribute to overall purpose and meaning.</p> <p><b>Reading for Informational Text 11.7-2.4</b> Determine the meaning of words and phrases as they are used in a text, including figurative, connotative, and technical meanings; analyze how an author uses and refines the meaning of a key term or phrase to shape a text's meaning, tone, and purpose.</p> <p><b>Reading for Informational Text 11.7-2.4</b> Integrate and evaluate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, and verbally) in order to make a case on a topic or issue, noting any discrepancies among the data.</p>                                                                                                                                                                                                                                                                                                                                                                  | <p><b>UNIT 1: Producting Your Success: TED Talk #1: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 1.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 1.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 1.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 1.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> |
| <p><b>UNIT 2: Producting Your Success: TED Talk #2: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 2.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 2.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 2.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 2.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> | <p><b>UNIT 2: Producting Your Success: TED Talk #2: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 2.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 2.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 2.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 2.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> |
| <p><b>UNIT 3: Producting Your Success: TED Talk #3: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 3.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 3.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 3.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 3.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> | <p><b>UNIT 3: Producting Your Success: TED Talk #3: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 3.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 3.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 3.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 3.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> |
| <p><b>UNIT 4: Producting Your Success: TED Talk #4: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 4.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 4.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 4.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 4.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> | <p><b>UNIT 4: Producting Your Success: TED Talk #4: The Power of Productive Failure</b> Submit a written response to Google Classroom.</p> <p><b>Unit 4.6 Team Activity</b> Divide into five seven teams. Each team is responsible for one of the sections of the extant business environment discussed in the chapter (business, technological, demographic, environmental, competitive, global, and technological). (Instructional Resources)</p> <p><b>Unit 4.7: Twenty core concepts from Unit 1 &amp; 2</b></p> <p><b>Critical Thinking Case Study: Uber-Hits a Bumpy Road</b></p> <p><b>Unit 4.8 &amp; Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Critical Thinking Case Study: We Want Our MTV Information</b></p> <p><b>Unit 4.8: Twenty Core Concepts on Unit 1 &amp; 2</b></p> <p><b>...Confirm Unit 3 &amp; 4 Team Activity</b> Divide the class into five teams. Each team is assigned a different section of the extant business environment to determine how it will help or hinder today's business. Each team will be required to prepare a presentation and a video to be developed by the lead developer. (Project: Information Technology)</p> <p><b>Preparing for Tomorrow's Workplace Skills:</b><br/>Do you have what it takes to be a successful franchisee?</p> <p><b>Working the Unit:</b><br/>Select three franchises that interest you.</p> |

Winter 2018 Faculty Course Feedback

**"How many quizzes are there in the entire course?"** Six: four on page 15, two on page 16.

# Retrieval and Navigation

When a question arrives, retrieval **narrows the document to a smaller set of pages or passages**, so the reasoner never has to process the whole thing.

- **Four ways to search.** By words, with a sparse lexical scorer. By learned similarity over text embeddings. By image, with late-interaction encoders that score the question against the rendered page directly. Or by structure, walking a tree or graph built over the document.
- **Structure is what similarity cannot do.** Scoring passages independently handles paraphrase well, but it cannot follow a cross-reference or a dependency that no single passage carries.
- **The main risk is omission.** If a page the answer requires never reaches the reasoner, nothing downstream can use what was on it.

Finding the evidence is only half the task. **The model still has to reason with it.**

# Reasoning Strategies

Once the evidence is assembled, the model still has to make sense of it. Some answers need **facts from different pages combined, step by step**.

- **Ask it to show its reasoning.** Chain-of-thought elicits the intermediate steps before the answer, which binds evidence from different pages one hop at a time and leaves a trace you can check.
- **Ask it to check itself.** Self-reflection has the model audit or revise an initial answer against the same evidence, optionally with a separate verifier.
- **Ask it several times.** Sampling multiple reasoning paths and taking the majority buys robustness against a single bad chain, and costs proportionally more compute.

All of it operates over a **fixed buffer of evidence assembled before the call**. None of it can recover evidence that was never supplied.

# Agentic Methods

An agentic system **decides what to inspect next as it goes**, rather than executing a plan fixed in advance. Each observation feeds the next decision.

- **It can go back for more.** Search again with a refined query, follow a cross-reference to another section, or request a closer look at a page it has already seen.
- **It can divide the work.** Some systems split the roles: a reader that gathers, a planner that decides, a checker that decides whether the answer is good enough.
- **Compute follows difficulty.** An easy question stops early; a hard one keeps going. The trajectory also becomes an audit trail of what the system actually looked at.

The trade-off is **time, and more opportunities for a bad decision to lead the system astray**. It also still inherits the recall ceiling of every tool it calls.

# Training Strategies

Almost every system starts from a model **already trained on text and images**. What gets trained after that is the document-specific machinery around it.

## Find

---

A retriever learns to surface better pages: aligning visual page embeddings with parsed text, folding in neighbouring-page context, or scoring pages by their role in a reasoning path rather than by similarity alone.

RETRIEVER

## Combine

---

A reasoner learns to produce structured reasoning on its own, by imitating demonstration traces and then by reinforcement against rewards that credit evidence recall as well as the final answer.

REASONER

## Decide

---

A controller learns when to inspect more and when to stop, usually by distilling a stronger model's trajectories first and then optimising the policy directly.

CONTROLLER

Each of those improvements depends on having suitable training examples. **That makes the available datasets a central part of the field.**

# Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

## Pretraining

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The exception rather than the norm. Genuinely multi-page corpora have appeared only recently, so most systems inherit a backbone pretrained on single pages and skip the stage entirely.

RARE

## Fine-tuning

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Where the weight falls. Broad instruction mixes of released question-answer collections, plus bespoke corpora synthesised by a strong model over scraped PDFs.

MOSTLY SYNTHETIC

## Benchmarking

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Accumulated in tiers, from single-page extraction through to long documents whose evidence spans pages. In practice evaluation has consolidated on a small, much-reused pool.

NARROW POOL

Benchmarks measure whether the answer was right. For this problem we also want to know **whether the right evidence was found and used** to get there.

# Why Locate the Failure?

The survey shows many ways to build these systems. But when a system answers wrongly, **different papers give different explanations for why**, and those explanations imply different fixes.

“How much did the leading region's revenue grow between the 2022 and 2023 tables?”

The system answers **14%**. The correct answer is 23%. What went wrong?

## The page was read badly

The parser garbled a figure in the table, or the page was rendered too small to read it.

## The right page was missed

Only one of the two tables was retrieved, so the comparison was never possible.

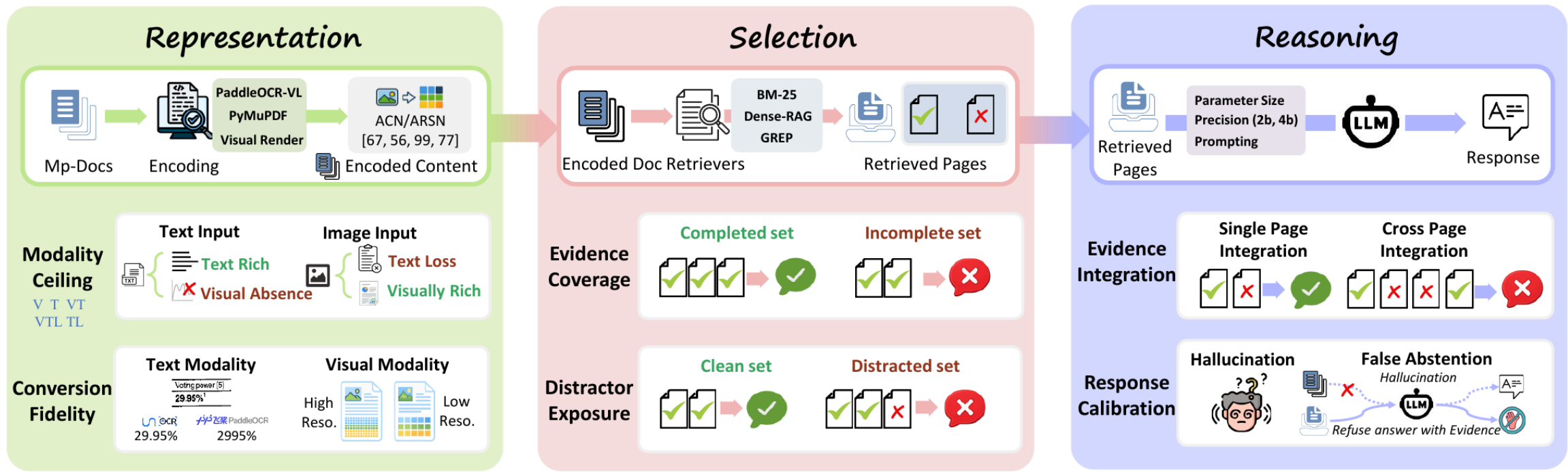
## The model could not use it

Both tables were present and legible, and it still picked the wrong row or did the arithmetic wrong.

End-to-end accuracy cannot tell these apart, and **under a fixed compute budget you have to know which one to spend on**.

# Three Failure Loci

Three places in the pipeline where an answer can go wrong, each with its own diagnostic question.



## REPRESENTATION

Did we preserve the information the question needs from the page?

## SELECTION

Did that information actually reach the model?

## REASONING

Did the model use it correctly once it had it?

The order matters, and it is what makes attribution possible: **we cannot reason over evidence that was lost in encoding, or never selected.** So a locus can only be tested once the ones above it are neutralised.

# How We Isolate a Failure

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

---

## Encode

Every page becomes text, an image, or both.  
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

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## Retrieve

▶ A retriever scores every encoded page against the question and returns the top  $k$ , restored to document order.

ColQwen3 · page-level

---

## Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass. Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Every reported accuracy carries a confidence interval** from a document-level bootstrap, because questions from one document are not independent.

# Representation Experiments

Two questions about the page itself: **what can an input express**, and **how much is lost converting the page into it**.

## Modality ceiling

WHAT THE INPUT CAN EXPRESS

---

Four rungs of increasing richness. **T** is the PDF's embedded text layer. **TL** is layout-aware parser markdown. **TLV** adds the rendered page image. **V** is the page image alone.

The comparison asks whether the required evidence is still recoverable once the textual, structural, or visual signal is taken away, rather than assuming one representation is best everywhere.

## Conversion fidelity

HOW MUCH SURVIVES THE CONVERSION

---

Hold the rung fixed and vary the quality of each channel instead: **the parser** for text, across PyMuPDF, MinerU, PaddleOCR-VL and Unlimited-OCR, and **the render resolution** for the image.

This separates what a modality cannot carry in principle from what a bad conversion destroyed in practice.

The annotated evidence pages are supplied in both, so **retrieval is not a variable here**. Whatever changes is the encoding.

# Selection Experiments

Selection can fail in two opposite ways, and the point of the experiment is that **they are not equally damaging**. Both start from the pages known to contain the answer.

## Evidence coverage

REQUIRED EVIDENCE REMOVED

---

Take questions that cite more than one gold page and **withhold one of them**. Pages are ranked within the document, so we can remove the highest-ranked gold page, the lowest-ranked, or keep only one.

A question that cannot satisfy a condition is excluded and recorded, never quietly satisfied with a different page set.

## Distractor exposure

IRRELEVANT EVIDENCE ADDED

---

Keep the complete gold set and **add up to five top-ranked non-gold pages** alongside it. Conditions are blocked on the exact gold-page count, so questions needing different amounts of evidence are never pooled.

The supplied context stays at six pages or fewer, so distractor count is not confounded with context length.

The encoding and the reasoner stay fixed across both, so **the two effects can be compared directly** against the same oracle baseline.

# Reasoning Experiments

With representation and selection controlled, what remains is the reasoner. Two different problems, so **they are tested separately**.

## Evidence integration

CAN IT COMBINE WHAT IT HAS?

---

Split the questions by the **structure of the evidence** they need: one page and one source, several sources on one page, or evidence distributed across pages.

Then add a step-by-step reasoning prompt and see which of those groups it actually helps. The annotated pages are supplied throughout, so any deficit arises after the evidence has arrived.

## Response calibration

SHOULD IT ANSWER AT ALL?

---

Vary both whether the evidence supports an answer and whether the model is **explicitly permitted to abstain**.

Answerable questions get their oracle gold pages. Unanswerable ones get the three top-ranked pages by BM25: plausible context that does not support an answer. That measures unsupported answers in one direction and false refusals in the other.

Everything is decoded greedily, so a condition is **reproducible from its identity** rather than from a seed.

# Datasets Make Attribution Possible

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

## MMLongBench-Doc

PRIMARY DIAGNOSTIC

|           |            |            |              |
|-----------|------------|------------|--------------|
| 135       | 47.5       | 847        | 244          |
| documents | mean pages | answerable | unanswerable |

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

## LongDocURL

SECONDARY REPLICATION

|           |            |           |              |
|-----------|------------|-----------|--------------|
| 396       | 85.6       | 2,325     | 8            |
| documents | mean pages | questions | unanswerable |

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

# Text and Vision Complement Each Other

With the correct pages supplied, accuracy climbs the representation ladder: **29.0% text only**, **42.6% image only**, **49.5% with both**.

| Evidence source | Accuracy    |             |             |             | Abstention  |            |            |            | Incorrect   |             |             |             |
|-----------------|-------------|-------------|-------------|-------------|-------------|------------|------------|------------|-------------|-------------|-------------|-------------|
|                 | T           | TL          | TLV         | V           | T           | TL         | TLV        | V          | T           | TL          | TLV         | V           |
| Text            | <b>36.9</b> | 44.4        | <b>51.0</b> | 42.3        | 23.0        | 11.5       | 7.3        | 8.0        | 40.1        | 44.1        | 41.6        | 49.7        |
| Layout          | 26.3        | 33.9        | 49.2        | <b>43.2</b> | 20.3        | 10.2       | <b>0.8</b> | <b>4.2</b> | <b>53.4</b> | <b>55.9</b> | <b>50.0</b> | 52.5        |
| Table           | 35.6        | <b>44.7</b> | 44.7        | 34.1        | <b>15.9</b> | <b>7.2</b> | 6.7        | 9.1        | 48.6        | 48.1        | 48.6        | <b>56.7</b> |
| Chart           | 16.3        | 16.9        | 42.7        | 40.4        | 32.6        | 28.1       | 7.3        | 6.2        | 51.1        | 55.1        | <b>50.0</b> | 53.4        |
| Figure          | 14.5        | 22.1        | 42.4        | 39.0        | 36.6        | 24.8       | 7.6        | 7.9        | 49.0        | 53.1        | <b>50.0</b> | 53.1        |
| All             | 29.0        | 35.7        | 49.5        | 42.6        | 26.3        | 17.2       | 6.2        | 7.1        | 44.7        | 47.2        | 44.3        | 50.3        |

Answer outcomes by evidence source and representation rung, MMLongBench-Doc, oracle pages

## – Text alone misses visual questions

At **T**, chart questions reach 16.3% and figure questions 14.5%. Adding the page image takes them to 42.7% and 42.4%.

## – It fails by refusing, not by guessing

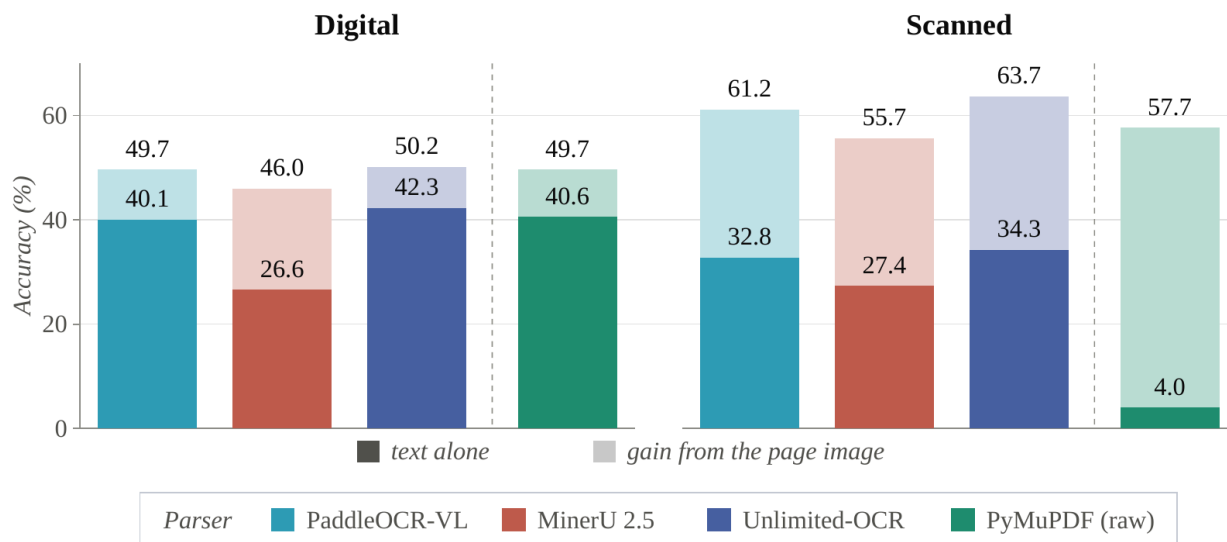
Abstention on figure questions falls from 36.6% to 7.6% across the ladder, while the incorrect rate stays broadly flat. The gain comes out of refusals.

## – Images alone are weaker than both

**V** reaches 42.6% against 49.5% for **TLV**. Text exposes fine detail in a form that is easier to use than recovering it from pixels.

# Conversion Quality Matters

A useful input type still fails if the page is converted badly. **Parser choice alone moves accuracy by tens of points**, so it is not neutral preprocessing.



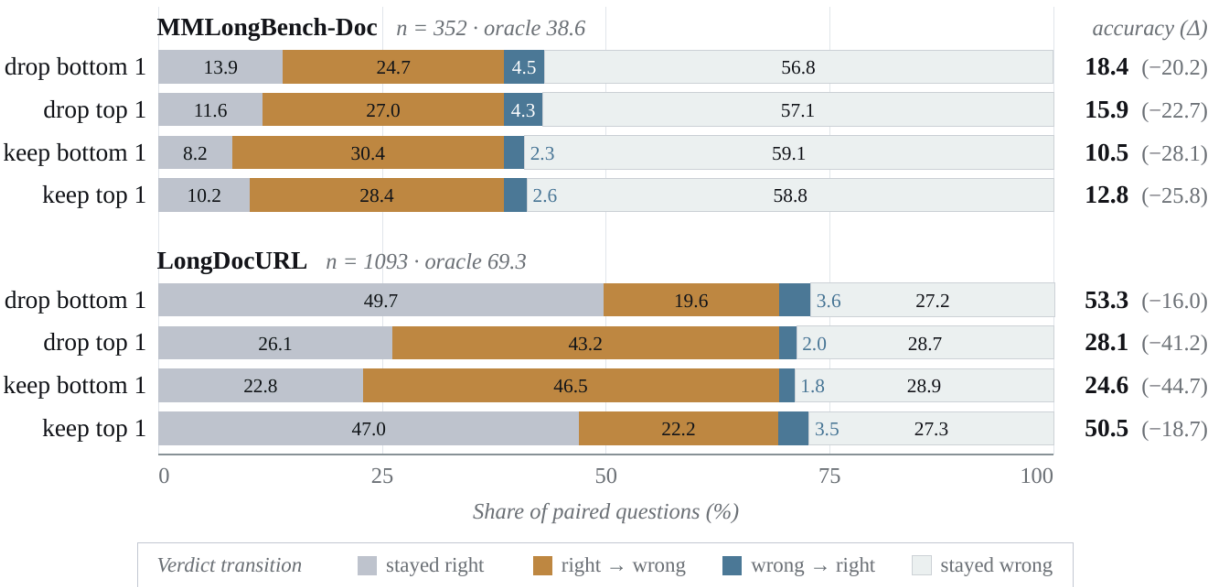
Parser accuracy by text source, with and without the page image

## READING IT

- **Scans break the text channel entirely.** On scanned documents the PDF's embedded text layer is often absent, and accuracy collapses. Conversion failure does not merely degrade the representation, it removes it.
- **Parsers differ a lot.** MinerU is the weakest on both digital and scanned pages; PaddleOCR-VL and Unlimited-OCR recover substantially more.
- **Vision covers for the parser.** Adding the page image compresses the spread between parsers sharply, and the weakest parser benefits most. Many conversion errors are recoverable once a second channel still holds the signal.
- **Resolution is the mirror image.** Raising render resolution helps **V** far more than **TLV**, where the text already carries part of what the extra pixels would recover.

# Missing Evidence Is Costly

Now remove one page the question needs. **Answers that were correct become wrong, and the reverse almost never happens.**



Verdict transitions after withholding or retaining gold evidence pages

## READING IT

- **Withholding one gold page breaks 24.7–27.0% of previously correct answers**, while only 4.3–4.5% are repaired. Keeping just one gold page widens it further: 28.4–30.4% break against 2.3–2.6% repaired.
- **The loss is directional.** This is more than the expected correlation between retrieval recall and accuracy. Removing known evidence produces a one-way loss that downstream reasoning rarely reverses.
- **Which page survives matters too.** Keeping the highest-ranked gold page preserves much more than keeping the lowest. Extractive questions tolerate partial evidence; counting and calculation do not.

# Moderate Distraction Is Less Damaging

The opposite error: retrieval returns the right pages **plus a few extra**. Within the range we test, this costs far less than missing a required page.

| Condition           | MMLongBench-Doc |      |      |      |      |      | LongDocURL |      |      |      |      |     |
|---------------------|-----------------|------|------|------|------|------|------------|------|------|------|------|-----|
|                     | T               | TL   | TLV  | V    | R→W  | W→R  | T          | TL   | TLV  | V    | R→W  | W→R |
| <i>1 gold page</i>  |                 |      |      |      |      |      |            |      |      |      |      |     |
| Oracle              | 37.1            | 44.6 | 63.7 | 55.0 | —    | —    | 65.6       | 74.0 | 74.7 | 64.5 | —    | —   |
| +1 distractor       | 39.5            | 48.9 | 63.7 | 55.7 | 7.8  | 7.0  | 65.4       | 72.4 | 74.4 | 63.0 | 8.5  | 8.1 |
| +2 distractors      | 38.8            | 47.9 | 64.3 | 56.1 | 8.2  | 8.0  | 65.1       | 68.7 | 72.0 | 63.9 | 9.3  | 7.5 |
| +3 distractors      | 40.9            | 47.0 | 63.1 | 55.1 | 9.7  | 8.2  | 64.5       | 68.0 | 68.4 | 62.1 | 11.7 | 6.9 |
| +4 distractors      | 41.9            | 45.3 | 62.5 | 52.5 | 9.5  | 6.6  | 63.5       | 68.0 | 72.2 | 62.5 | 10.5 | 8.7 |
| +5 distractors      | 39.5            | 42.6 | 62.3 | 54.4 | 10.0 | 6.9  | 64.6       | 69.1 | 70.2 | 60.8 | 11.4 | 7.3 |
| <i>2 gold pages</i> |                 |      |      |      |      |      |            |      |      |      |      |     |
| Oracle              | 28.9            | 33.3 | 37.8 | 34.6 | —    | —    | 62.2       | 63.4 | 70.4 | 61.5 | —    | —   |
| +1 distractor       | 25.7            | 34.0 | 38.6 | 34.4 | 7.8  | 7.5  | 58.7       | 63.0 | 66.2 | 57.9 | 9.6  | 4.1 |
| +2 distractors      | 25.3            | 33.6 | 35.7 | 32.4 | 9.6  | 6.0  | 58.9       | 62.2 | 66.4 | 57.4 | 10.9 | 4.4 |
| +3 distractors      | 21.6            | 28.2 | 33.6 | 33.2 | 11.4 | 6.0  | 58.1       | 63.3 | 70.8 | 57.8 | 9.7  | 6.0 |
| +4 distractors      | 24.6            | 30.7 | 43.7 | 31.3 | 11.8 | 11.8 | 57.8       | 63.2 | 68.3 | 57.1 | 13.3 | 6.1 |

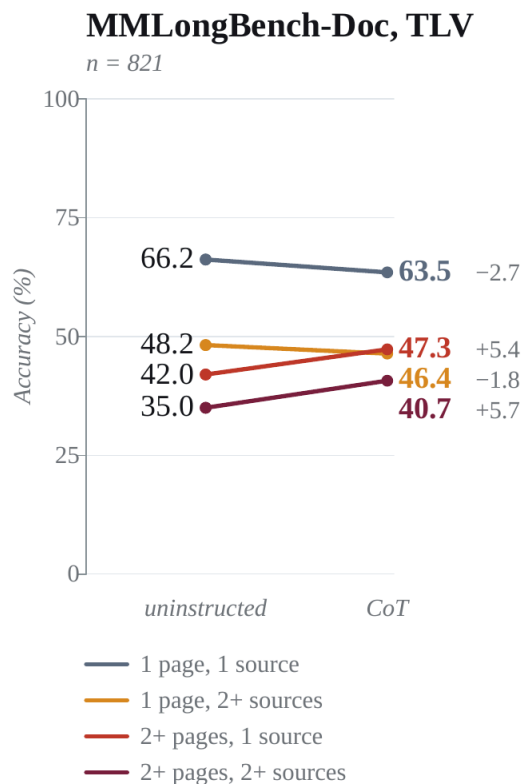
A complete gold set, then the same set plus one to five distractor pages. Tested range only: context stays at six pages or fewer

## READING IT

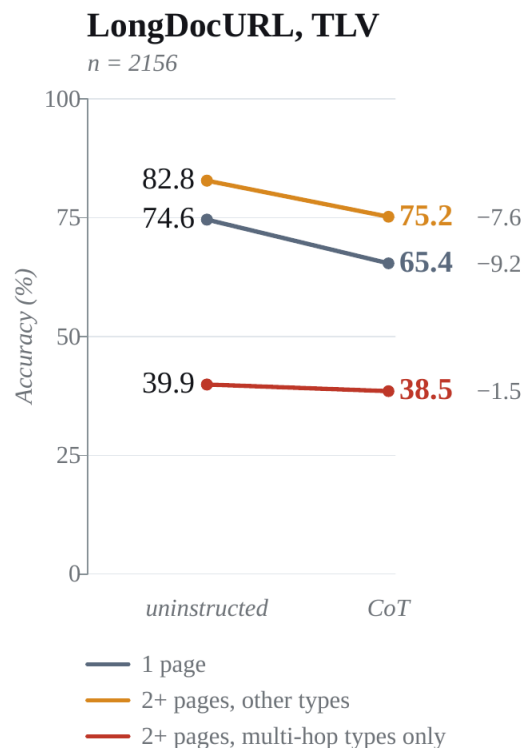
- **Aggregate accuracy barely moves.** Single-page questions stay broadly stable, and multi-page questions decline gradually. LongDocURL degrades even less under comparable distractor counts.
- **But stability is not harmlessness.** The verdict transitions show added pages both breaking and repairing answers. Positive and negative effects partly cancel at the aggregate level.
- **A non-gold page is not always irrelevant.** Sometimes it adds a partial instance that makes the model overcount; sometimes it supplies the context that disambiguates the gold pages and repairs the answer.
- **So lean toward coverage** when the later context can absorb it, and control how the retrieved evidence is presented rather than minimising how much of it there is.

# Reasoning Across Pages Remains Hard

Every correct page is supplied, in the richest representation. **Cross-page questions are still much harder**, and asking for step-by-step reasoning helps only some of them.



Effect of chain-of-thought across evidence structures, TLV rung, oracle evidence pages

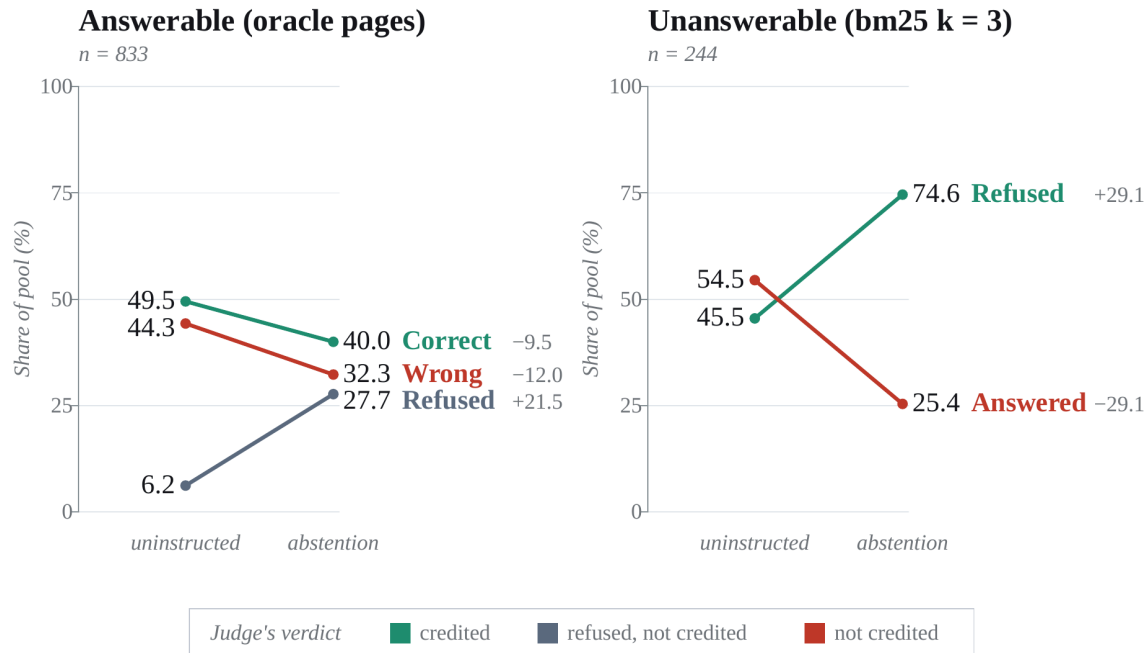


## READING IT

- **The deficit is real.** On MMLongBench-Doc, one-source questions fall from 66.2% on a single page to 42.0% once the evidence spans pages, and to 35.0% when several sources must be combined as well.
- **Chain-of-thought helps precisely there.** It slightly hurts single-page questions (66.2 to 63.5), and lifts cross-page ones (42.0 to **47.3**, and 35.0 to **40.7**). The pooled integration deficit narrows from 24.6 to 18.2 points.
- **Most of the gain is structured evidence.** Tables improve by **10.9** points; charts, text and layout by two to three; figures fall slightly.
- **And it does not generalise.** On LongDocURL the same prompt reduces accuracy almost everywhere. More reasoning helps when it matches the task, not because there is more of it.

# Abstention Is Not Well Calibrated

The last reasoning decision is whether the evidence supports an answer at all. An abstention instruction makes the model more cautious, **but not more discerning**.



Effect of targeted abstention prompting at the TLV rung

## READING IT

- **It works where it should.** On unanswerable questions, refusal rises from 45.5% to **74.6%**, cutting unsupported answers substantially.
- **And it fires where it should not.** On answerable questions, refusal rises from 6.2% to **27.7%**, and correct answers fall from 49.5% to 40.0%.
- **Difficulty is being read as insufficiency.** Under the instruction, refusal reaches 35.5% for figure questions and 30.3% for tables, against 24.1% for text, with oracle pages supplied, so the evidence is definitely there.
- **The prompt moves willingness, not judgement.** It shifts the operating point rather than separating sufficient evidence from insufficient.

# A Simple Adaptive Method

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

## REPRESENTATION

### Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

## SELECTION

### Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

## REASONING

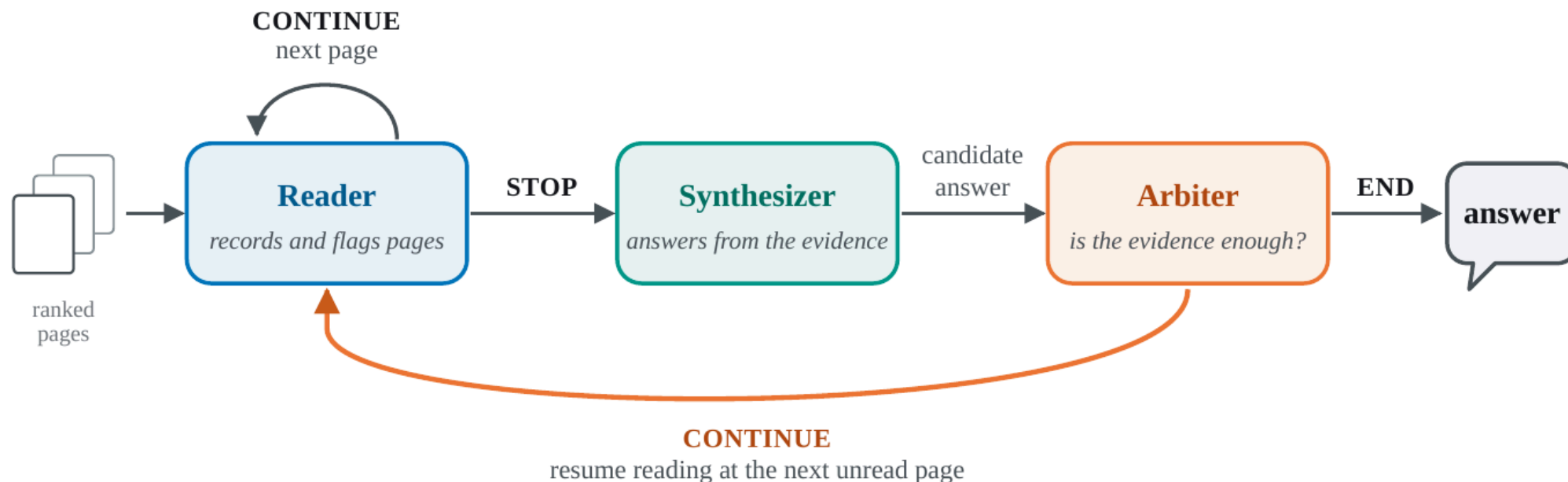
### Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

# Reader, Synthesizer, Arbiter

One underlying model in **three roles**, differentiated only by the instruction and the input each receives.



## reader

Works through the pages in retrieval-ranked order, recording answer-relevant evidence as it goes, and proposes stopping when it thinks it has enough.

## synthesizer

Receives the **most important pages in full**, plus the short records from everything else read, and proposes an answer from that.

## arbiter

Accepts the answer, or sends the reader back for more. It cannot write an answer itself, which makes **stopping too early recoverable** rather than final.

The loop is what makes the amount of reading **adapt to the question**: an easy one stops early, a hard one keeps going.

# System-Level Validation, and Its Cost

On MMLongBench-Doc the method reaches **54.5%** under the official scorer, above the gold-page baseline using the same model family.

| Method                                                   | OCR | Trained        | Evidence source |             |             |             |             | Evidence pages |             |             | ACC         |
|----------------------------------------------------------|-----|----------------|-----------------|-------------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|
|                                                          |     |                | TXT             | LAY         | CHA         | TAB         | FIG         | SIN            | MUL         | UNA         |             |
| Official protocol: answer extraction + rule-based scorer |     |                |                 |             |             |             |             |                |             |             |             |
| Gold pages + VLM (TLV)                                   | ✓   | ×              | 48.1            | 49.6        | 39.4        | 50.5        | 42.5        | 57.7           | 37.4        | 52.5        | 49.5        |
| <i>Qwen2.5-VL-7B</i>                                     |     |                |                 |             |             |             |             |                |             |             |             |
| MDocAgent <sup>†</sup>                                   | ✓   | ×              | 43.1            | 31.2        | 32.6        | 38.7        | 37.9        | 53.5           | 23.8        | 28.3        | 38.5        |
| MoLoRAG+                                                 | ×   | ✓ <sup>a</sup> | 42.7            | 38.8        | 33.3        | 38.5        | 40.7        | 52.9           | 27.6        | 35.9        | 41.0        |
| MM-Doc-R1 <sup>†</sup>                                   | ✓   | ✓              | 46.2            | 38.1        | 40.8        | 52.8        | 35.9        | 56.0           | 31.2        | 68.2        | 49.7        |
| <i>Qwen3-VL-8B</i>                                       |     |                |                 |             |             |             |             |                |             |             |             |
| Whole document                                           | ×   | ×              | —               | —           | —           | —           | —           | 46.9           | 35.3        | 25.8        | 38.5        |
| MARDoc                                                   | ✓   | ×              | —               | —           | —           | —           | —           | —              | —           | —           | 52.7        |
| DocTrace                                                 | ✓   | ✓              | —               | —           | —           | —           | —           | 53.2           | <b>41.1</b> | 70.5        | 52.9        |
| Ours (4-bit)                                             | ✓   | ×              | <b>48.0</b>     | <b>41.9</b> | <b>43.7</b> | <b>59.9</b> | <b>41.8</b> | <b>58.9</b>    | 37.8        | <b>71.9</b> | <b>54.5</b> |

MMLongBench-Doc accuracy, official protocol. Published systems on open 7-8B backbones, values taken from the cited papers

## READING IT

- **Above the oracle baseline.** **54.5%** against **49.5%** for the same model reading only the annotated gold pages, which suggests useful context comes from **beyond** the pages the benchmark labels as evidence.
- **And above the published comparisons** on the same backbone, the strongest being 52.9%.
- **The gain is not uniform.** Multi-page questions reach 37.8%, still below DocTrace's 41.1%. Cross-page integration remains the hard part, exactly as the attribution study predicted.

8.2 GiB

mean peak VRAM

13.9

agent turns per question

9.3 min

median latency per question

16.6 min

mean, almost all decoding

# Conclusion

Long-document understanding is, above all, a matter of **managing evidence well**.

---

## Preserve

What survives the conversion of a page into text, pixels, and structure.

---

## ► Select

Which of it actually reaches the model, out of hundreds of pages.

---

## ► Reason

Whether the model combines it, and knows when it is not enough.

---

### 1 The survey

maps how systems preserve, find, and use information across pages.

### 2 The empirical study

shows where the failures arise, and that each one needs a different fix.

### 3 The simple method

shows those lessons can be turned back into a working system.

**Thank you, and happy to take questions.**